University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

7,029: Quality Management for Superior Performance and Innovation

ECTS credits: 3

Overview examination/s

(binding regulations see below)
decentral - Oral examination and technical discussions, Analog, Individual work individual grade (100%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
7,029,1.00 Quality Management for Superior Performance and Innovation -- English -- Budde Lukas

Course information

Prerequisites

This course is assigned to the profile "Supply Chain & Operations Management" but may be taken without selecting a major.

Learning objectives

- The students understand the concept of holistic quality management, as a business management and implementation task (TQM- Total Quality Management).
- Students will understand how to use quality management as a competitive factor.
- Students understand how quality management methods can be used to determine the quality of products or services.
- Students understand how to create a quality plan (while considering e.g. financial and sustainable aspects).
- Students can use quality management methods to measure and improve the quality of products, services or processes.
- Students understand the relationship between innovation and quality management.
- Students will be able to apply preventive quality management methods (e.g. FMEA).

Content

Quality Management belongs to these kind of topics which are relevant for every kind of company, either the company is producing and selling products (e.g. food, cars or smartphones) or providing services (e.g. flights, accommodation, etc.). This is why quality management is an integral part of almost every company. Although quality is mainly associated with preventing failures of products (e.g. car or machine breakdowns), the quality management in a company extends of many different functional areas in a company (from planning, operations and till optimization). It is an integral part of the optimization in the company, offering better products or services fulfilling better customer demands and thus providing a competitive advantage for the company. Beside the customer focus it is also a driver of social responsibility. Quality is thus a central driver for innovation in a company. Thus, it is not a surprise that newer concepts of quality management make more and more use of available data of products, production and supply chain and customer usage to increase the quality.

This course conveys students practical knowledge about concepts, steps and processes of Quality Management. Thereby tasks of the quality management will be illustrated along physical products and digital service examples.

Exemplary questions addressed in this course:

- How much quality is needed to fulfill customer demands (e.g. how much power should the car be)?
- How can we measure the quality of our services (e.g. check-in-services at the airport)?
- How to make use of data to optimize products (e.g. which features of the notebook are used?)
- How to detect and prevent failures in production?

These and other questions will be addressed during the lecture and the exercises along with illustrative practical examples.

Structure and indications of the learning and teaching design

- This course comprises 3 credits. Accordingly, the total workload for students is 90 hours. This includes self-study, attendance time and the examination.
- The structure of the contact study is planned as follows: 1.5 hours of lecture time every week
- The structure of the self-study is planned as follows: 24 hours of preparation time for the course and 30 hours of preparation time for the examination performance
- The course is conducted in physical format.

Starting from this semester, there is one lecture unit per week of 1.5 h each. In total there will be 8 lectures and 4 exercises to train and apply the methods and approaches presented in the lectures.

This means concretely for this course:

- LE1: Quality Management as the competitive factor - Introduction
- LE2: Quality Planning for products and services
- LE3: **Exercise** Quality Planning - House of Quality
- LE4: Environmental and social responsibility
- LE5: **Exercise** House of Quality - Design of a quality plan
- LE6: Quality Assurance and Control of operations
- LE7: Quality Testing
- LE8: **Exercise** Quality Measurement
- LE9: Quality Optimization based on data - Insights from Practice
- LE10: **Exercise** Optimization
- LE11: Summary
- LE12: Exam

Literature

Friedli, T., Basu, P. K., Mänder, C., Calnan, N., Biehl, S., Braun, M., Buess, P., Köhler, S., Lamba, S. S., Lehmann, B., Lembke, N., Lima, J. M., Lupo, M., Moloney, C., Ponce, N., Reddy, M. M., & Schneider, U. (2018). *21c Quality Management in the Pharmaceutical Industry*. Editio Cantor Verlag.

Iwaarden, J. van, & Wiele, A. van der. (2006). *Innovative quality management cases*. Emerald Group Publishing.

Kanji, G. K. & Sheffield Hallam University Centre for Quality and Innovation. (1996). *Total quality management in action* (1st ed. 1996.). Chapman & Hall. https://doi.org/10.1007/978-94-009-1543-5

Karmarkar, U. S., & Lederer, P. J. (1997). *The practice of quality management* (1st ed. 1997.). Kluwer Academic Publishers. https://doi.org/10.1007/978-1-4615-6283-2

Naidu, N. V. R. (2006). *Total quality management*. New Age International P Ltd., Publishers.

Seghezzi, H. D., & Seghezzi, H. D. (2007). *Integriertes Qualitätsmanagement: Der St. Galler Ansatz* (3., vollst. überarb. Aufl.). Hanser.

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/1)

Examination modalities

Examination type	Oral examination and technical discussions
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	100%
Duration	--

Languages
Question language: English
Answer language: English

Remark
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Examination-aid rule
Closed Book

The use of aids is prohibited as a matter of principle, with the exception of pocket calculator models of the Texas Instruments TI-30 series and, in case of non-language exams, bilingual dictionaries without any handwritten notes. Any other aids that are admissible must be explicitly listed by faculty members in the paragraph entitled “Supplementary aids” of the course and examination fact sheet; this list is exhaustive.

Procuring any aids, as well as ensuring their working order, is the exclusive responsibility of students.

Supplementary aids

The use of aids is limited; any additional aids permitted are exhaustively listed under "Supplementary aids". Basically, the following is applicable: At such examinations, all the pocket calculators of the Texas Instruments TI-30 series and mono- or bilingual dictionaries (no subject-specific dictionaries) without hand-written notes are admissible. Any other pocket calculator models and any electronic dictionaries are inadmissible. In addition, any type of communication, as well as any electronic devices that can be programmed and are capable of communication such as notebooks, tablets, mobile telephones and others, are inadmissible. Students are themselves responsible for the procurement of examination aids.

Content

Oral exam to the topics of:

- LE1: Quality Management as the competitive factor - Introduction
- LE2: Quality Planning for products and services
- LE3: **Exercise** Quality Planning - House of Quality
- LE4: Environmental and social responsibility
- LE5: **Exercise** House of Quality - Design of a quality plan
- LE6: Quality Assurance and Control of operations
- LE7: Quality Testing
- LE8: **Exercise** Quality Measurement
- LE9: Quality Optimization based on data - Insights from Practice
- LE10: **Exercise** Optimization
- LE11: Summary

Literature

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Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).